9th Class Physics – Past Paper 2021

Punjab Board (Rawalpindi Board)

Group-II (Paper Code: 5474)

Objective Type

Marks: 12
Time: 15 Minutes

Multiple Choice Questions (MCQs)

Q.No.	Question	A	B	C	D
1	Latent heat of vaporization value	2.26×10^5 JKg^{-1}	**2.26×10^6 JKg^{-1}** ✓	2.26×10^6 Nm	2.26×10^6 J
2	In solids, heat is transferred by	Absorption	**Conduction** ✓	Radiation	Convection
3	Heat from fireplace reaches us by	Absorption	**Radiation** ✓	Convection	Absorption
4	200 μs equivalent	0.2 s	0.02 s	**2×10^{-4} s** ✓	2×10^{-6} s
5	Amount of substance is measured in	Gram	Kilogram	Newton	**Mole** ✓
6	Displacement ÷ Time =	Speed	Acceleration	**Velocity** ✓	Deceleration
7	Change in position is called	Speed	Velocity	**Displacement** ✓	Distance
8	Unit of momentum	Nm	Kgm s^{-2}	**Ns** ✓	Ns^{-1}
9	Second condition of equilibrium	ΣFx = 0	ΣFy = 0	ΣF = 0	**$\Sigma\tau = 0$** ✓
10	Value of gravitational constant (G)	**6.673×10^{-11} Nm^2Kg^{-2}** ✓	6.6×10^{-11}	9.1×10^{-31}	1.6×10^{-19}
11	Unit of energy	Watt	**Joule** ✓	Newton	Mole
12	One Pascal equals	**1 Nm^{-2}** ✓	10^4 Nm^{-2}	10^2 Nm^{-2}	10^3 Nm^{-2}

Subjective Type

Marks: 48
Time: 1 Hour 45 Minutes

Section – I

Question 2

Write short answers of any FIVE.

1. Define prefixes and give an example.
2. What is meant by vernier constant?
3. Why is rolling friction less than sliding friction?
4. Define vectors and scalars.
5. What is International System of Units (SI)?
6. Define position and give an example.
7. Define momentum and write its unit.
8. Can a body moving at constant speed have acceleration? Justify.

Question 3

Write short answers of any FIVE.

1. Write two uses of artificial satellites.
2. Define axis of rotation.
3. Write the value of G and its SI unit.
4. Define potential energy and write its equation.
5. Define centre of gravity.
6. What is the difference between like and unlike parallel forces?
7. Define efficiency and write its formula.
8. How can the mass of Earth be determined?

Question 4

Write short answers of any FIVE.

1. Define pressure and write its SI unit.
2. Why does conduction of heat not take place in gases?
3. Define tensile strain. Why does it have no unit?

4. How does heat reach us from the Sun?
5. Define heat and temperature.
6. Define specific heat and write its SI unit.
7. State Hooke's Law. What is meant by elastic limit?
8. What is evaporation? On which factors does the evaporation of a liquid depend?

Section – II

Attempt any TWO questions.

Question 5

(a)

Derive with the help of a graph that:

$$2aS = V_f^2 - V_i^2$$

(4 Marks)

(b)

How much time is required to change a momentum of **22 Ns** by a force of **20 N**?

(5 Marks)

Question 6

(a)

Define equilibrium. State and explain the first condition of equilibrium.

(4 Marks)

(b)

A cyclist does **12 Joules** of useful work while pedalling his bike from every **100 Joules** of food energy.

Calculate his efficiency in percentage.

(5 Marks)

Question 7

(a)

State Pascal's Law and explain the hydraulic press.

(4 Marks)

(b)

A container has **2.5 litres of water at 20°C**.

How much heat is required to boil the water?

(5 Marks)

End of Paper